**//Shortest Path**

```
#include<stdio.h>
#include<stdlib.h>
#define MAX_NODES 10
#define INFINITY 10000
//ENUMERATION FOR MAKING NODE PERMANENT
enum node
{
permanent,tentative
};
int path[MAX_NODES],dist[MAX_NODES][MAX_NODES];
int src,dst,n,count=0;
//NODE STRUCTURE
struct state { int
predecessor; int
length;
enum node lbl;
}state[MAX_NODES];
void shortestpath(int ,int );
void main() {
int i,j ;
printf("\n\n\t\tSHORTEST PATH CALCULATION \n");
printf("\nEnter the number of nodes(LESS THAN 10): ");
scanf("%d",&n); for(i=0;i<n;i++)  for(j=0;j<n;j++)
dist[i][j]=0; for(i=0;i<n;i++)
{
 for(j=0;j<n;j++)
 {  if((dist[i][j]==0) &&
(i!=j))
 {  printf("Enter the distance between %d & %d :
",i+1,j+1);  scanf("%d",&dist[i][j]);  dist[j][i]=dist[i][j];
 }
 } }
printf("\nDistance Matrix\n"); for(i=0;i<n;i++)
{  for(j=0;j<n;j++)
printf("\t%d",dist[i][j]);
printf("\n"); }
printf("\nEnter source node: ");
scanf("%d",&src); printf("\nEnter
destination node:");
scanf("%d",&dst);
if(src==dst) {
```

```
 printf("\nThere is no path for self\n");
exit(0); }
shortestpath(src-1,dst-1); printf("\nThe
shortest path is:"); while(count>0)
{  if(count==1)
printf("%d",path[count-1]+1);
 else
 printf("%d==>",path[count-1]+1);  count--; }
printf("\n\nDistance of shortest path is: %d\n",state[dst-1].length);
}
void shortestpath(int s,int d)
{ int
i,k,min,j;
struct state *p;
for(p=&state[0];p<&state[n];p++)
{  p->predecessor=-1;
p->length=INFINITY;
p->lbl=tentative;
}
//MAKE SOURCE NODE PERMANENT
state[s].length=0;
state[s].lbl=permanent;
k=s; do {
 for(i=0;i<n;i++)
 {  if(dist[k][i]!=0 &&
state[i].lbl==tentative)
 {  if((state[k].length+dist[k][i]) <
state[i].length)
 {
state[i].predecessor=k;
 state[i].length=state[k].length+dist[k][i];
 }
 }
 }
 k=0;
 min=INFINITY;
for(i=0;i<n;i++)  {
if(state[i].lbl==tentative &&
(state[i].length< min))
 {
min=state[i].length;
 k=i;
}
```

```
 }
state[k].lbl=permanent;
}while(k!=d); i=0; k=d;
do {  path[i++]=k;
k=state[k].predecessor;
count++;
}while(k>=0);
}
```